BALAJI ABBU

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Summary

Electrical Design Engineer (B.Tech EEE) experienced in preparing SLDs, electrical layouts, cable schedules, BOMs and technical documentation in MS Excel and AutoCAD, supported by design calculations. Solid knowledge of LV distribution, power systems and earthing, with a disciplined approach to IEC-compliant, quality-checked deliverables.

Professional Experience

- **Thermopads Private Limited, Hyderabad** — On-site
  Graduate Engineer Trainee - Electrical Design — *June 2026 – Present*
  - **Responsibilities**: Support engineering and design of industrial Electrical Heat Tracing (EHT) systems across the project lifecycle: customer enquiry, technical specification review, design, documentation and execution.
    • Maintain project engineering documents and participate in design reviews and technical meetings with project, engineering and marketing teams.
    • Prepare engineering documentation and design deliverables for EHT projects, including: SLDs, Electrical Layouts, Wiring Diagrams, Design Schedules, Cable Schedules, Bill of Quantities (BOQ), Bill of Materials (BOM), Data Sheets, Quality Assurance Plan (QAP), General Arrangement (GA) Drawings, CRS Reports, FAT and SAT Reports, Sub-Vendor Lists, Master Document Lists, Sample Heat Loss Calculation Reports, Cable Voltage Drop Calculation Sheets, Technical documentation and project records.
- **Siri Power Systems Technologies Pvt Ltd, Hyderabad** — On-site
  Electrical Maintenance Intern — *6 Months, Dec 2022 – May 2023*
  - **Scope**: For a period of 6 months, assisted in preventive maintenance of electrical systems and achieved a 15% reduction in equipment downtime.
  - **Outcomes**: Gained hands-on experience with electrical tools and equipment including multimeters, megohmmeters, clamp meters and testers.

Technical Skills

- **Electrical Design**: SLDs, Electrical Layouts, Wiring Diagrams, GA Drawings, Cable Schedules, Cable Sizing, Cable Voltage Drop Calculations, BOM/BOQ, Datasheets, QAP, FAT/SAT Documentation.
- **Software Tools**: ETAP (Basic), AutoCAD Electrical, MATLAB, Simulink, PVsyst, DIALux evo, Proteus, Arduino IDE, MS Office.
- **Concepts**: Power Systems, LV Distribution, Transformers, Switchgear, Circuit Breakers (MCCB/ACB), UPS, PDUs, Earthing, Lightning Protection, Protection Systems, Power Electronics, Electrical Machines, TIA-942, IEC and IEEE Standards.
- **Soft Skills**: Cross-team Coordination, Analytical Thinking, Attention to Detail, Technical Communication, Time Management.

Education

- **Narsimha Reddy Engineering College** — Hyderabad, Telangana
  B.Tech - Electrical and Electronics Engineering; CGPA: 8.52 — *September 2023 - April 2026*
- **St. Peter's Engineering College** — Hyderabad, Telangana
  Diploma - Electrical and Electronics Engineering; CGPA: 8.59 — *October 2020 - April 2023*

Projects

- **ML-Based Driver Drowsiness and Alcohol Detection with Engine Locking System** – ***Python, OpenCV, MediaPipe, Arduino IDE***: Developed a road safety system combining ML-based real-time drowsiness detection, threshold-based alcohol sensing and automatic engine locking, with keypad authentication and buzzer alerts, so the vehicle runs only under safe conditions.
- **3-Ph LVAC Transmission and Distribution System** – ***Power Systems, Electrical Wiring***: Designed a real-time working model of a 3-phase LVAC transmission and distribution system, covering design, simulation and performance analysis, and collaborated with the team to prepare reports and deliver presentations.
- **Automated Plant Irrigation System** – ***Arduino UNO, Electronics***: Developed an automated plant irrigation system using sensors and a microcontroller to monitor soil moisture levels and control water supply automatically. The system activates the water pump when the soil becomes dry and stops irrigation when sufficient moisture is detected. This project helps conserve water and reduce manual effort.

Certifications And Workshops

- **Certifications**: MATLAB Onramp, Simulink Onramp, Simscape Onramp, Circuit Simulation Onramp, Introduction to Electric Vehicles, Basics of Python.
- **Workshops**: 2-Day Workshop on Embedded Systems, 2-Day Workshop on PCB Design.